

SMART UTM — a tensile rig that runs, watches and stops its own tests

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One specimen now yields a CURVE, not a point.

12 automation features and 6 closed-loop test protocols, every one validated on the rig. Camera-based strain sits INSIDE the control loop, so the machine measures the specimen instead of measuring itself.

12 / 16

smart features
built & validated

6

closed-loop test
protocols

17

runs auto-indexed,
13 specimens

± 1.3 N

force hold over
877 s

26 μe

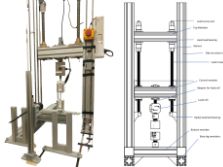
DIC noise floor
(900 s window)

0.9 %

agreement between two
fracture protocols

THE MACHINE

The rig — built in-house, every control loop closed in software



Two steppers (1.85 Nm, 20:1) drive 5 mm lead screws against a fixed base. An ANYLOAD 3 t cell measures force; a Basler camera measures strain optically at 20.8 px/mm from two markers on the 80 mm gauge. The ESP32 runs STEP/DIR/ENABLE only and knows nothing of test protocols — every protocol, guard and measurement here lives in the PyQt6 app, closing the loop at ~11 Hz.

A conventional pull gives ONE modulus, ONE yield, ONE strength — and a human presses every button. Crosshead travel is not specimen strain, so the machine reports its own compliance as material behaviour. A destructive test is one irreversible shot. And nothing watches the run, so a stall, a lost marker or a fracture are all noticed late, by a person.

HOW IT WORKS

One control engine

All six protocols are one function called on every load sample (~11 Hz) with load, position and time-matched DIC strain in hand. A protocol only returns a speed — it cannot bypass the shared safety net.

Strain from the specimen

A camera tracks two markers; engineering strain is $(L-L_0)/L_0$ at 20.8 px/mm. The centroid resolves 0.1 px steps at ± 0.02 px.

Simulated before hardware

Every policy is replayed against a spring plant and recorded CSVs — 9/9 checks pass before a specimen is risked.

WHAT IT BUYS YOU

The machine reads **STIFFER** while the specimen **SOFTENS**



Progressive cyclic to fracture. Crosshead stiffness ROSE 744 → 777 N/mm as rig slack squeezed out, while the DIC-measured specimen modulus FELL 2.32 → 1.73 GPa, a 26 % loss. Read from the crosshead alone this test says the material is getting stiffer — the strongest argument for putting the camera in the loop.

LIMITATIONS

Poisson's ratio is not measurable today

The gauge is too narrow: the elastic width change is sub-pixel at 20.8 px/mm. Every stress here is ENGINEERING (F/A_0) , never 'true'.

No fatigue damage curve yet

One run lost the camera mid-way and its repeat started on an already-damaged specimen, and the modulus RECOVERED between them — so no valid E_0 exists.

Creep and fatigue are each $n = 1$

Both measurements are proven; the material constants are not. Creep needs 3–4 stress levels and fatigue needs repeats.

The motor's torque ceiling is variable

Normally 3.2–3.4 kN, but ~2.6 kN on one session. The drivetrain is rated far above that, so the cause is electrical — not the software.

THE SMART FEATURES, SF1 TO SF16

■ built & rig-validated (12)

■ planned (3)

■ hardware-blocked (1)

SF1 Live DIC health HUD — 2 / 2

A green/amber/red badge shows marker lock, tracking % and jitter live — so a dropout is seen during the test, not found later in the CSV

SF3 Recipes / settings — 1 click

Saves and restores an entire setup, including all six protocols' parameters — so repeats are identical by construction, not by memory

SF5 Auto-stop at fracture — 1.2 s

A load-collapse detector halts the motor the moment the specimen breaks

SF7 Stall guard — 0.05 mm / 6 s

Halts a motor that has stopped moving under load

SF9 Six closed-loop protocols — 6 / 6

cyclic · staircase · relaxation · creep · staircase → FRACTURE · progressive-cyclic → FRACTURE

SF11 Auto-metadata + foldering — planned

Per-specimen folders and file IDs filled from the recipe, plus a one-click per-specimen slide deck

SF13 Guided workflow + live overlay — planned

A Connect → Calibrate → Mount → Prepare → Run → Save wizard, with live modulus and a fracture flag as the test runs

SF15 Test registry — 17 runs

Each run indexed with whatever its loading path supports — UTS, ϵ_f , E , σ_y , toughness — plus the force anchor that turns a tared reading into a true load

SF2 Prepare specimen — 4 → 1

One button tares position, force and DIC and clears the plots

SF4 One-click report — 1 click

A one-page PDF and every plot, recomputed from the CSV by the same library the app runs live

SF6 Constant-strain-rate testing — 0.00051 / s

Closed loop on the specimen's own de/dt : the crosshead adapted itself 0.10 → 0.05 mm/s, fast in the elastic region, slow through necking

SF8 Release load — ≤ 5 N

Taring at preload makes the readout say 0 N while 300 N is still on the specimen

SF10 Auto-preload — 1.03 ×

Approaches on a 0.2 → 0.1 → 0.02 mm/s schedule and stops 3 % high, because PLA relaxes about 2 % while held

SF12 DIC auto-calibrate — planned

Exposure and threshold swept on camera start, and an ROI that follows the markers through a ductile draw

SF14 Poisson's ratio / true stress — blocked

The four-marker maths is written and self-tested

SF16 Dead-DIC guard + hard backstops — 518 → 175 N

A controller ramping blind on a stale camera reading spikes the force